🔬 Photovoltaic Dataset Noise & Data Quality Analysis Report

Dataset: `PV_Diagnostic_Final_Labels.csv` (226,175 records)
Target Columns: `P1` (Panel 1 Power), `P2` (Panel 2 Power), `Irr` (Irradiance), `timestamp`

1. Executive Summary

This report provides a granular audit of data corruption, noise, and calculation anomalies present in the PV diagnostic dataset.

A critical finding is that **11.19% of records in Panel 1** and **8.71% in Panel 2** contain corrupted power values resulting from:

1. **Polarity / Sign Inversion ($P = -I \times V$)**: Caused by sensor polarity or datalogger sign inversion.
2. **Extreme Physical Outliers**: Values violating theoretical maximum solar generation ($P_{\text{max}} = 147.25\text{ W}$).
3. **Calculation Mismatches**: Unexplained arbitrary values diverging from $I \times V$.
4. **Night-time / Low-Irradiance artifacts**: Distortions during zero-sunlight hours.

2. Panel 1 Power (P_1) Noise Breakdown

Theoretical physical bounds based on maximum panel ratings ($I_{1,\text{max}} = 7.75\text{ A}$, $V_{1,\text{max}} = 19.00\text{ V}$): $$\mathbf{P_{1,\text{physical_max}} = 7.75\text{ A} \times 19.00\text{ V} = 147.25\text{ W}}$$

P_1 Quality Statistics (Total Rows: 226,175)

Category / Anomaly Type	Row Count	Percentage	Minimum Value	Maximum Value	Physical Impact
✅ **Perfectly Correct ($P_1 = I_1 \times V_1$)**	**200,874**	**88.81%**	0.00 W	146.95 W	Healthy physical data
❌ **Inverted Sign ($-I_1 \times V_1$)**	**24,533**	**10.85%**	-473.67 W	-0.0001 W	Negative power generation (Sensor bug)
❌ **Calculation Mismatch / Glitch**	**768**	**0.34%**	-0.91 W	334.72 W	Uncorrelated corrupted readings
** ⚠️ Extreme Outliers ($	P_1	> 147.25\text{ W}$)**	**711**	**0.31%**	-473.67 W
🔴 **Total Corrupted Rows in P_1**	**25,301**	**11.19%**	**-473.67 W**	**334.72 W**	**Destroys LSTM & ML loss convergence**

P_1 Noise Distribution Across Fault Classes

Fault Class (`Label_S1`)	Total Rows	Correct ($P_1 = I_1 \times V_1$)	Sign Inverted ($-I_1 \times V_1$)	Mismatch / Glitch	Noise Percentage
Degradation	103,885	102,105 (98.29%)	1,780 (1.71%)	0 (0.00%)	**1.71%**
Normal	47,303	43,794 (92.58%)	2,770 (5.86%)	739 (1.56%)	**7.42%**
Open Circuit	58,171	42,245 (72.62%)	15,926 (27.38%)	0 (0.00%)	**27.38%**
Shading	16,763	12,716 (75.86%)	4,047 (24.14%)	0 (0.00%)	**24.14%**
Short Circuit	53	14 (26.42%)	10 (18.87%)	29 (54.72%)	**73.58%**
Total	**226,175**	**200,874 (88.81%)**	**24,533 (10.85%)**	**768 (0.34%)**	**11.19%**

> ⚠️ **CAUTION:** In the **`Open Circuit`** and **`Shading`** categories, over **24% to 27%** of all data points have inverted negative power signs. In **`Short Circuit`**, **73.58%** of samples are corrupted!

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3. Panel 2 Power (P_2) Noise Breakdown

Theoretical physical bounds ($I_{2,\text{max}} = 7.75\text{ A}$, $V_{2,\text{max}} = 18.97\text{ V}$):
$$\mathbf{P_{2,\text{physical_max}} = 7.75\text{ A} \times 18.97\text{ V} = 147.02\text{ W}}$$

P_2 Quality Statistics (Total Rows: 226,175)

Category / Anomaly Type	Row Count	Percentage	Minimum Value	Maximum Value	Physical Impact
✅ **Perfectly Correct ($P_2 = I_2 \times V_2$)**	**206,470**	**91.29%**	0.00 W	146.95 W	Healthy physical data
❌ **Inverted Sign ($-I_2 \times V_2$)**	**19,619**	**8.67%**	-6.28 W	-0.0001 W	Negative power generation
❌ **Calculation Mismatch / Glitch**	**86**	**0.04%**	-0.0084 W	232.15 W	Corrupted sensor reading
** ⚠️ Extreme Outliers ($	P_2	> 147.02\text{ W}$)**	**26**	**0.01%**	150.12 W
🔴 **Total Corrupted Rows in P_2**	**19,705**	**8.71%**	**-6.28 W**	**232.15 W**	Corrupted signal

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P_2 Noise Distribution Across Fault Classes

Fault Class (`Label_S2`)	Total Rows	Correct ($P_2 = I_2 \times V_2$)	Sign Inverted ($-I_2 \times V_2$)	Mismatch / Glitch	Noise Percentage
Degradation	105,672	104,118 (98.53%)	1,554 (1.47%)	0 (0.00%)	**1.47%**
Normal	51,864	49,361 (95.17%)	2,441 (4.71%)	62 (0.12%)	**4.83%**
Open Circuit	52,104	37,673 (72.30%)	14,431 (27.70%)	0 (0.00%)	**27.70%**
Shading	16,402	15,216 (92.77%)	1,186 (7.23%)	0 (0.00%)	**7.23%**
Short Circuit	133	102 (76.69%)	7 (5.26%)	24 (18.05%)	**23.31%**
Total	**226,175**	**206,470 (91.29%)**	**19,619 (8.67%)**	**86 (0.04%)**	**8.71%**

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4. Environmental & Temporal Signal Noise

A. Low Irradiance & Night-Time Artifacts ($Irr < 50\text{ W/m}^2$)

- **Total Rows**: **4,985 rows (2.20%)**
- **Issue**: During night/early morning hours, $I \approx 0$ and $P \approx 0$ irrespective of panel health.
- **Breakdown by Label**:
  - Shading: 4,093 rows (82.1%)
  - Normal: 783 rows (15.7%)
  - Degradation: 90 rows (1.8%)
  - Short Circuit: 17 rows (0.3%)
  - Open Circuit: 2 rows (0.1%)

B. Time Discontinuities (Gaps)

- **Short Gaps (> 5 seconds)**: 56 occurrences
- **Long Night Gaps (> 1 hour)**: 11 occurrences (representing day-night transitions)
- **Risk**: Sliding windows (e.g. sequences of 30 or 60 steps) that cross these gaps combine non-contiguous physical time periods.

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5. Concrete Solution & Cleansing Script

To eliminate **100% of the power calculation noise**, execute the following preprocessing pipeline before training any Machine Learning or LSTM model:

```
import pandas as pd
import numpy as np

def clean_pv_dataset(csv_path="PV_Diagnostic_Final_Labels.csv"):
    df = pd.read_csv(csv_path, sep=";")
    df['timestamp'] = pd.to_datetime(df['timestamp'])

    # 1. Fix Power Noise programmatically (P = I * V)
    df['P1'] = df['I1'] * df['V1']
    df['P2'] = df['I2'] * df['V2']

    # 2. Filter out dark / night periods
    df_clean = df[df['Irr'] >= 50].sort_values('timestamp').reset_index(drop=True)

    # 3. Validation Check
    assert (df_clean['P1'] < 0).sum() == 0, "Error: Negative P1 detected!"
    assert (df_clean['P2'] < 0).sum() == 0, "Error: Negative P2 detected!"
    assert (df_clean['P1'] > 150).sum() == 0, "Error: Outlier P1 detected!"

    print(f"Data Cleansed Successfully! Remaining valid rows: {len(df_clean):,}")
    return df_clean

if __name__ == "__main__":
    df_clean = clean_pv_dataset()
```